Web Programming Course
Exam Simulation – April 16th, 2025

- This exam is open-book, but all work must be done individually.
- Answer each question clearly, using sketches, tables or descriptions where necessary.
- Provide examples of requests and responses to show how they work.
- Do not write actual code, but provide detailed API designs, request/response formats, and data models.

We have been asked to design a system for handling events. The system must provide:
- An API to create a new event (with event ID, title, description, date, location);
- An API to display the list of events in the system;
- An API to display the information of the individual event;
- An API to register for an event as a participant, indicating the user's first and last name.

Note that the event must be uniquely identified by its ID, while it can have the other fields not unique. It can be assumed that there is only one event per day, so a string with the date can be used as the ID, and it is automatically created by the system (not passed by the user during creation). The event resource can be used in different APIs. A simplified system can be assumed whereby users are not required to authenticate.

Question 1: API Definition (15 points)

- Define API endpoints (URL structures and HTTP methods); (2 points for each API)
- Specify request parameters and possible responses, possibly with an example of a successful request; (1 point for each API)
- Detail the expected data models for requests and responses. (3 points total)

Question 2: Error Handling (5 punti)

A good API must handle errors effectively.
- Describe common error responses (e.g., 400, 404, 500) and when they should be used in the context of the newly designed API. (3 points)
- Provide response examples for at least two error cases. (1 point for each example)

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Solution

A proposed solution is presented below. Note that this presents only a design cue, but should not be understood as the only possible solution.

Question 1: API Definition

Endpoint:

- POST /events → Creates an event.
- GET /events → Visualizes the list of events.
- GET /events/{id} → Visualizes a single event.
- POST /events/{id}/register → Used to subscribe to an event.

In the various APIs, the data type Event will be used, composed as follows:

- ID: string
- Title: string
- Description: string
- Date: string
- Location: string

Example:

POST /events
```
{
  "title": "Hackathon 2025",
  "description": "Programming event.",
  "date": "2025-06-20",
  "location": "Milan"
}
```
Response:
```
{
  "id": "20250620",
  "message": "Event successfully created"
}
```

This API allows the creation of a new event. The user will send in the body the information about the event, i.e., all the fields of the Event object defined above except the ID, which is assigned by the system. The server will respond with the event id.

GET /events

```
[
  {
    "id": 101,
    "title": "Hackathon 2025",
    "description": "Programming event.",
    "date": "2025-06-20",
    "location": "Milan"
  }
]
```

This API returns a list of Event objects, listing previously entered events.

GET /events/20250620

```
  {
    "id": "20250620",
    "title": "Hackathon 2025",
    "description": " Programming event.",
    "date": "2025-06-20",
    "location": "Milan"
  }
```

This API returns information regarding a specific Event object, identified by its ID via path parameter.

POST /events/20250620/register

```
  {
    "name": "Mario",
    "surname": "Rossi",
    "email": "mario.rossi@example.com"
  }
```

Response: 200 OK

This API allows a user to enroll in one of the events, identified by its ID via a path parameter. The user submits the first name, last name and email and receives in response, if successful, the response code 200.

Question 2: Error Handling

Error responses are used by the server to notify the user of different types of errors encountered during the execution of requests.

- **Error 400.** This error characterizes an incorrect request by the user. For example, in the case of the newly designed API, it could be returned if the user sends an invalid email address in the body of the POST request for registration, or if he/she sends a request with an empty body.
- **Error 404.** This error is used to tell the user that not only is the request incorrect, but that the requested resource does not exist. It can be returned by APIs to non-existing endpoints (e.g., "/event list/{id}"), or even in case the user specifies a non-existing {id} parameter in the event list.
- **Error 500.** This error is used to indicate a server-side error. For example, it may be returned by the API for entering a new event if the server is unable to save the information sent by the user. Another example would be if the list of events becomes too large to handle and causes the server to crash inadvertently.